



US012400497B2

(12) **United States Patent**
Laurila et al.

(10) **Patent No.:** **US 12,400,497 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **ACCESS CONTROL SYSTEM AND A METHOD FOR CONTROLLING OPERATION OF AN ACCESS CONTROL SYSTEM**

(71) Applicant: **KONE Corporation**, Helsinki (FI)

(72) Inventors: **Jussi Laurila**, Helsinki (FI); **Visa Rauta**, Helsinki (FI)

(73) Assignee: **KONE Corporation**, Helsinki (FI)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 121 days.

(21) Appl. No.: **17/939,146**

(22) Filed: **Sep. 7, 2022**

(65) **Prior Publication Data**

US 2023/0010991 A1 Jan. 12, 2023

Related U.S. Application Data

(63) Continuation of application No. PCT/FI2020/050245, filed on Apr. 15, 2020.

(51) **Int. Cl.**
G07C 9/00 (2020.01)
G07C 9/15 (2020.01)
G07C 9/37 (2020.01)

(52) **U.S. Cl.**
CPC **G07C 9/00182** (2013.01); **G07C 9/15** (2020.01); **G07C 9/37** (2020.01)

(58) **Field of Classification Search**
CPC **G07C 9/00182**; **G07C 9/37**; **G07C 9/15**
USPC **340/5.7**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2013/0298466 A1* 11/2013 Osann, Jr. E05G 5/02 49/98
2017/0053435 A1* 2/2017 Reymann G06V 20/52
2017/0103265 A1* 4/2017 Channah G07C 9/37
2018/0293523 A1 10/2018 Bergdale et al.
2020/0081418 A1* 3/2020 Dyne G06Q 10/04
2022/0228419 A1* 7/2022 Tiso G07C 9/253

FOREIGN PATENT DOCUMENTS

KR 10-2016-0091666 A 8/2016

OTHER PUBLICATIONS

Chinese Office Action and Search Report for Chinese Application No. 202080099692.3, dated Nov. 27, 2023, with English translation. International Search Report (PCT/ISA/210) issued in PCT/FI2020/050245, dated Dec. 18, 2020. Written Opinion of the International Searching Authority (PCT/ISA/237) issued in PCT/FI2020/050245, dated Dec. 18, 2020.

* cited by examiner

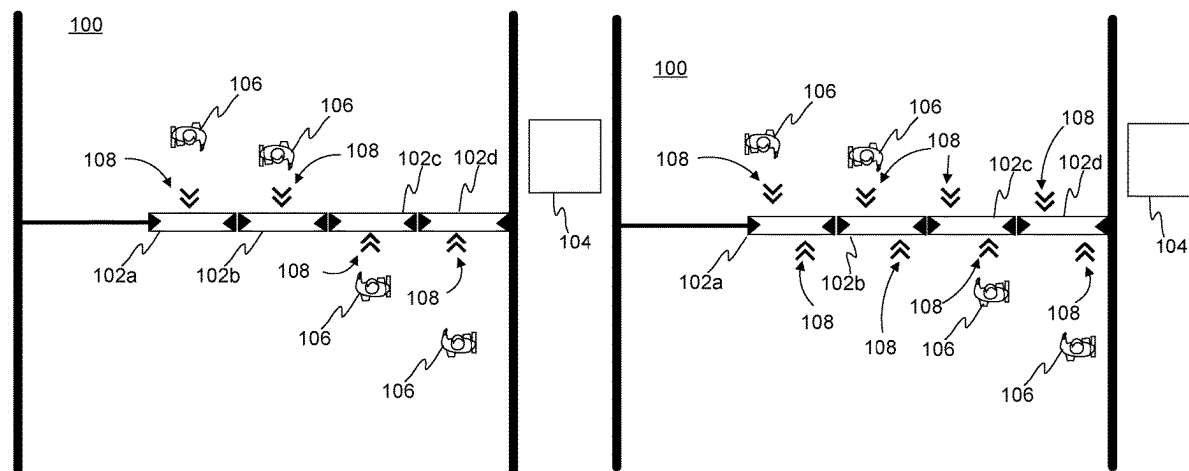
Primary Examiner — Kam Wan Ma

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch & Birch, LLP

(57) **ABSTRACT**

An access control system includes: at least one gate device and a control unit. The control unit is configured to: control the at least one gate device to operate in a first operation mode, in which the gate device is arranged to allow access for the users via the gate device only from a first access direction, detect that a traffic flow of the users meets a predefined capacity limit, and control at least one gate device to change to operate in a second operation mode, in which the gate device is arranged to allow access for the users via the gate device from the first access direction and from the second access direction. The invention relates also to a method for controlling an access control system.

20 Claims, 3 Drawing Sheets



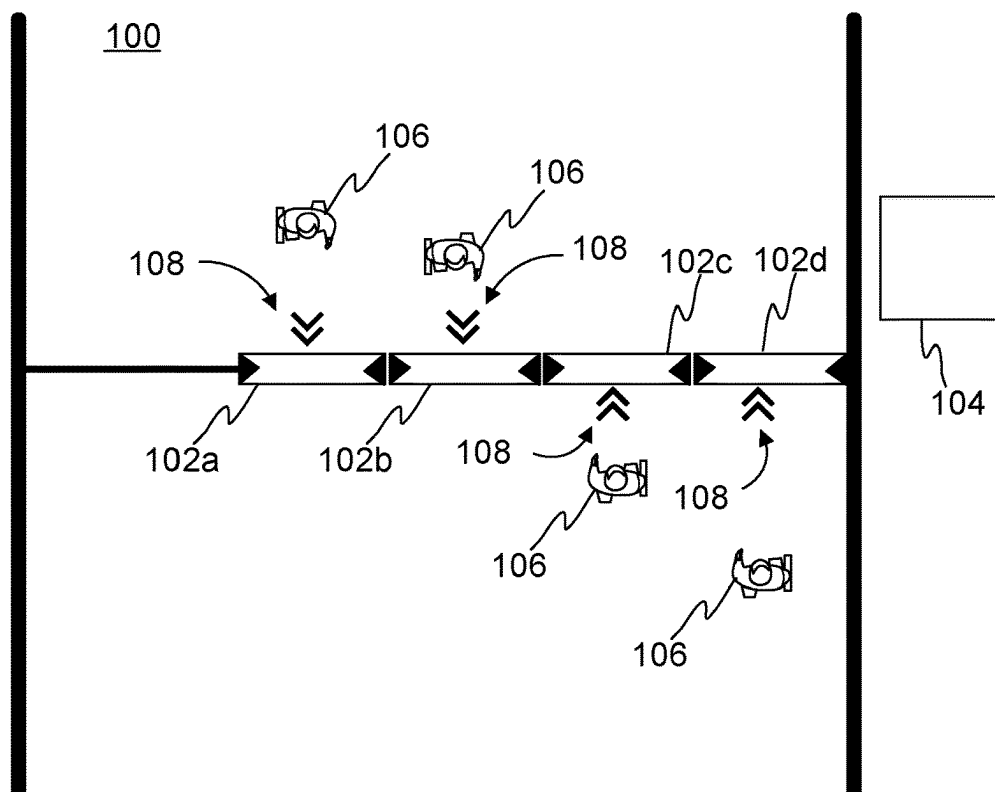


FIG. 1A

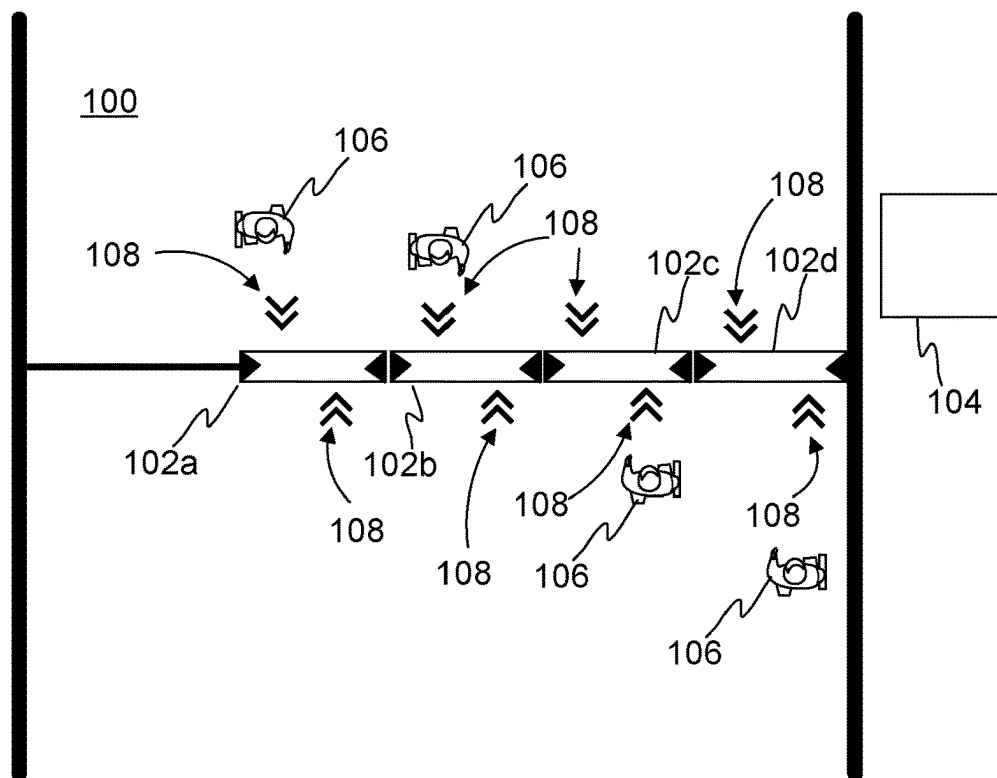


FIG. 1B

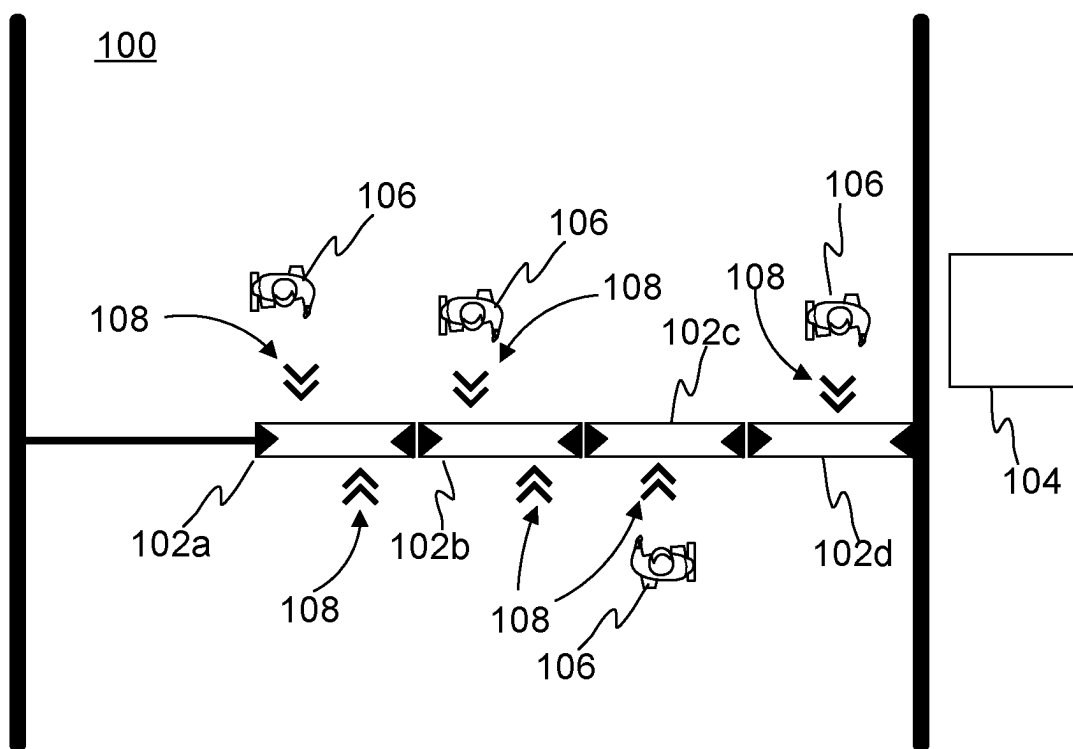


FIG. 1C

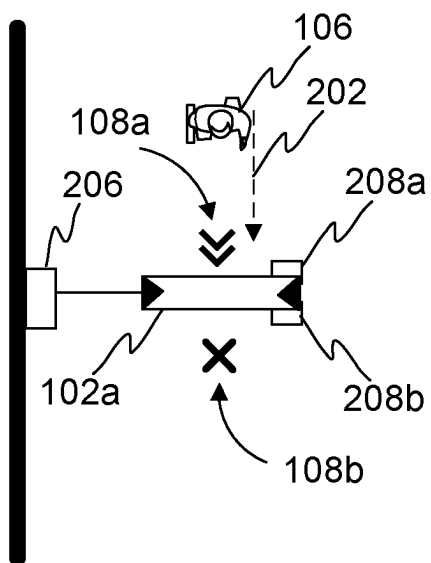


FIG. 2A

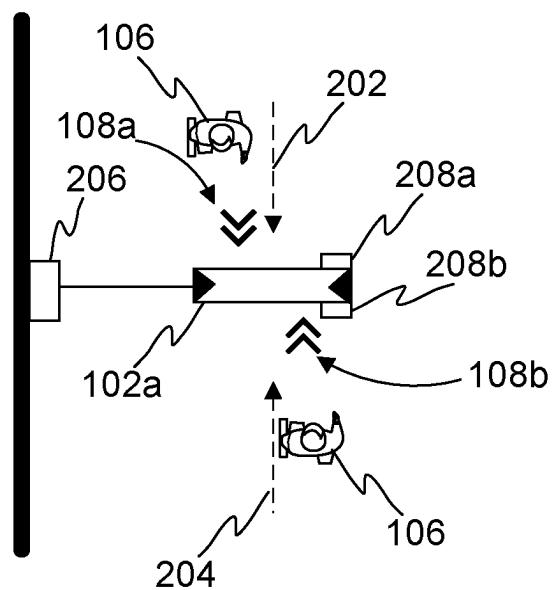


FIG. 2B

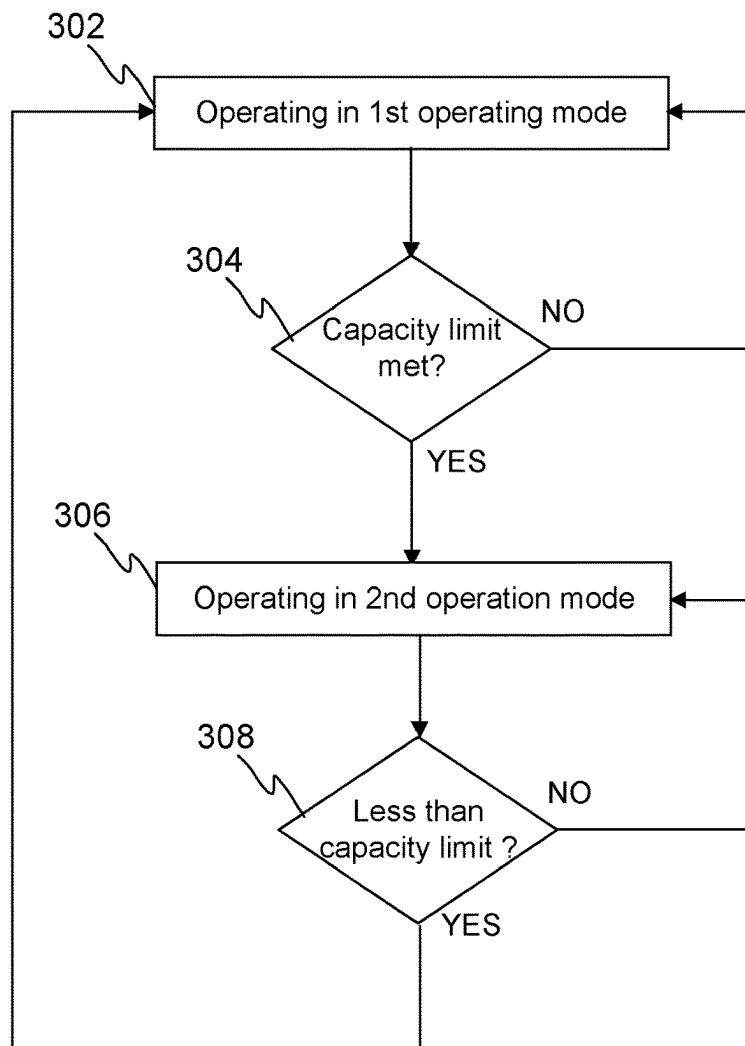


FIG. 3

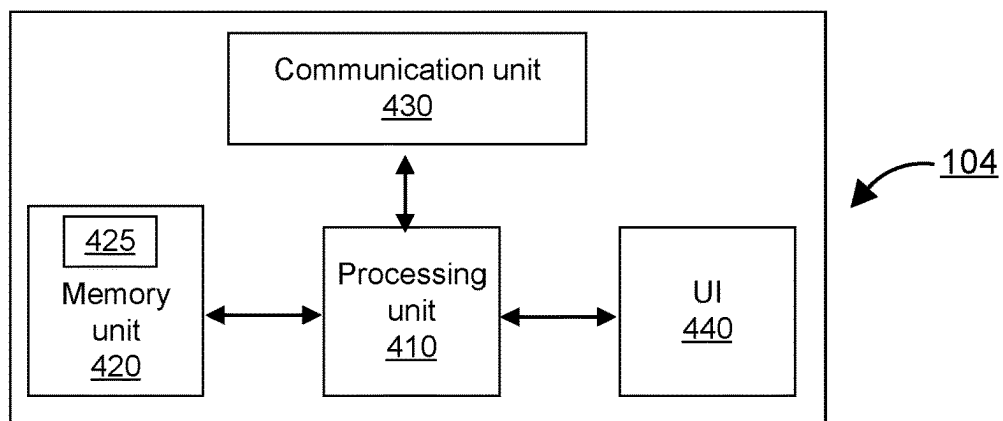


FIG. 4

1

ACCESS CONTROL SYSTEM AND A METHOD FOR CONTROLLING OPERATION OF AN ACCESS CONTROL SYSTEM

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of PCT International Application PCT/FI2020/050245, filed on Apr. 15, 2020, which is hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

The invention concerns in general the technical field of access control. Especially the invention concerns access control systems.

BACKGROUND

Typically, gate devices, such as security gates and turnstiles, may comprise access control. The access control enables that only authorized users may have access through the gate device. The access control may be based on using keycards; tags; identification codes, such as personal identity number (PIN) code, ID number; and/or biometric technologies, such as fingerprint, facial recognition, iris recognition, retinal scan, voice recognition, etc.

The gate device allows one-way access through the gate device. Typically, two or more gate devices may be arranged next to each other so that at least one gate device allows access through the gate device from a first direction and at least one gate device allows access through the gate device from a second direction being opposite to the first direction.

In high traffic situations, e.g. during rush hours, the capacity of the gate device may not be enough, i.e. the number users that the gate device allows to access through the gate device may be too low. This may cause lines and jamming of the gate device.

Thus, there is need to develop further solutions in order to improve the operation of the gate devices.

SUMMARY

The following presents a simplified summary in order to provide basic understanding of some aspects of various invention embodiments. The summary is not an extensive overview of the invention. It is neither intended to identify key or critical elements of the invention nor to delineate the scope of the invention. The following summary merely presents some concepts of the invention in a simplified form as a prelude to a more detailed description of exemplifying embodiments of the invention.

An objective of the invention is to present an access control system and a method for controlling operation of an access control system. Another objective of the invention is that the access control system and the method for controlling operation of an access control system improve an operation capacity of one or more gate devices of the access control system.

The objectives of the invention are reached by an access control system and a method as defined by the respective independent claims.

According to a first aspect, an access control system is provided, wherein the access control system comprises: at least one gate device and a control unit configured to: control the at least one gate device to operate in a first operation

2

mode, in which the gate device is arranged to allow access for the users via the gate device only from a first access direction; detect that a traffic flow of the users meets a predefined capacity limit; and control at least one gate device to change to operate in a second operation mode, in which the gate device is arranged to allow access for the users via the gate device from the first access direction and from the second access direction.

The access control system may further comprise at least one imaging device configured to obtain image data from the first access direction and from the second access direction, wherein the control unit may be configured to: receive the obtained image data from the first access direction in the first operation mode of the gate device or from the first access direction and the second access direction in the second operation mode of the gate device, identify authorized users approaching the at least one gate device based on the received image data and prestored data associated with the authorized users by using facial recognition-based identification, and control the at least one gate device to allow the access only for the authorized users via the at least one gate device.

Alternatively or in addition, the control unit may further be configured to: detect that the traffic flow of the users is less than the predefined capacity limit and control the at least one gate device to change back to operate in the first operation mode.

In the second operation mode the gate device may be arranged to allow simultaneous access of the users via the gate device from the first access direction and from the second access direction.

The capacity limit of the at least one gate device may be defined by a predefined number of users accessing via the at least one gate device per a predefined time period.

Alternatively or in addition, the access control system may further comprise at least one indication device arranged to provide a visual indication indicating the first access direction and/or the second access direction depending on the operation mode of the gate device.

Moreover, in the second operation mode the at least one indication device may be arranged to provide visual indication indicating two lanes, wherein one of the two lanes may indicate the first access direction and the other of the two lanes may indicate the second access direction.

Alternatively or in addition, the at least one indication device may comprise one or more displays on which the visual indication may be displayed and/or one or more or projectors configured to project the visual indication on a surface in a vicinity of the at least one gate device.

According to a second aspect, a method for controlling operation of an access control system is provided, wherein the access control system comprises at least one gate device allowing access for users via the at least one gate device and a control unit, wherein the method comprises: controlling, by the control unit, the at least one gate device to operate in a first operation mode, in which the gate device is arranged to allow access for the users via the gate device only from a first access direction; detecting, by the control unit, that a traffic flow of the users meets a predefined capacity limit; and controlling, by the control unit, at least one gate device to change to operate in a second operation mode, in which the gate device is arranged to allow access for the users via the gate device from the first access direction and from a second access direction.

The system may further comprise at least one imaging device configured to obtain image data from the first access direction and from the second access direction, wherein the

3

method may further comprise: receiving, by the control unit, the image data from the first access direction in the first operation mode of the gate device or from the first access direction and the second access direction in the second operation mode of the gate device; identifying, by the control unit, authorized users approaching the at least one gate device based on the received image data and prestored data associated with the authorized users by using facial recognition-based identification; and controlling, by the control unit, the at least one gate device to allow the access only for the authorized users via the at least one gate device.

Alternatively or in addition, the method may further comprise: detecting, by the control unit, that the traffic flow of the users is less than the predefined capacity limit; and controlling, by the control unit, the at least one gate device to change back to operate in the first operation mode.

In the second operation mode the gate device may be arranged to allow simultaneous access of the users via the gate device from the first access direction and from the second access direction.

The capacity limit of the at least one gate device may be defined by a predefined number of users accessing via the at least one gate device per a predefined time period.

Alternatively or in addition, the method may further comprise providing a visual indication indicating the first access direction and/or the second access direction depending on the operation mode of the gate device.

Moreover, the method may further comprise providing in the second operation mode a visual indication indicating two lanes, wherein one of the two lanes may indicate the first access direction and the other of the two lanes may indicate the second access direction.

Alternatively or in addition, the visual indication may be displayed on one or more displays and/or projected on a surface in a vicinity of the at least one gate device.

Various exemplifying and non-limiting embodiments of the invention both as to constructions and to methods of operation, together with additional objects and advantages thereof, will be best understood from the following description of specific exemplifying and non-limiting embodiments when read in connection with the accompanying drawings.

The verbs “to comprise” and “to include” are used in this document as open limitations that neither exclude nor require the existence of unrecited features. The features recited in dependent claims are mutually freely combinable unless otherwise explicitly stated. Furthermore, it is to be understood that the use of “a” or “an”, i.e. a singular form, throughout this document does not exclude a plurality.

BRIEF DESCRIPTION OF FIGURES

The embodiments of the invention are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

FIGS. 1A-1C illustrate schematically an example of an access control system according to the invention.

FIG. 2A an example of operation of a gate device according to the invention in a first operation mode.

FIG. 2B an example of operation of a gate device according to the invention in a second operation mode.

FIG. 3 illustrates schematically an example of a method according to the invention.

FIG. 4 illustrates schematically an example of components of a control unit according to the invention.

DESCRIPTION OF THE EXEMPLIFYING EMBODIMENTS

FIGS. 1A-1C illustrate schematically an example of an access control system 100 according to the invention. The

4

access control system 100 comprises at least one gate device 102a-102d allowing access for users 106 via the at least one gate devices 102a-102d and a control unit 104. In the example of FIGS. 1A-1C the access control system 100 comprises four gate devices 102a-102d arranged next to each other. However, the invention is not limited to that and the access control system 100 may comprise any other number of gate devices 102a-102d.

In the example of FIG. 1A the gate devices 102a-102d are configured to operate in a first operation mode. In the first operation mode of the gate device 102a-102d the gate device 102a-102d is arranged to allow access for the users 106 via the gate device 102a-102d only from one access direction. In other words, in the first operation mode the gate device 102a-102d allows one-way access via the gate device 102a-102d.

In the example of FIG. 1B the gate devices 102a-102d are configured to operate in a second operation mode. In the second operation mode of the gate device 102a-102d the gate device 102a-102d is arranged to allow access for the users 106 via the gate device 102a-102d from both access directions. In other words, in the second operation mode the gate device 102a-102d allow two-way access via the gate device 102a-102d. In the example of FIG. 1B all gate devices 102a-102d are configured to operate in the second operation mode.

In the example of FIG. 1C the gate devices 102c and 102d are configured to operate in the first operation mode in which the gate devices 102c and 102d are arranged to allow access for the users 106 via the gate device 102c and 102d only from one access direction, and the gate devices 102a and 102b are configured to operate in the second operation mode, in which the gate devices 102a and 102b are arranged to allow access for the users 106 via the gate devices 102a and 102b from both access directions.

The access control system 100 may further comprise at least one indication device (not shown in FIGS. 1A-1C) arranged to provide visual indication 108 indicating the access direction(s) of the gate devices 102a-102d depending on the operation mode of the gate devices 102a-102d. The at least one indication device may comprise one or more displays on which the visual indication 108 may be displayed and/or one or more projectors configured to project the visual indication 108 on a surface in a vicinity of, i.e. close to, the gate device 102a-102d as illustrated in the example of FIGS. 1A-1C, wherein the visual indications 108 are projected on a surface, e.g. a floor, in front of the gate devices 102a-102a in the access direction(s) of the gate devices 102a-102d.

The control unit 104 is configured to control the operations of the access control system 100. The control unit 104 may be external entity or it may be implemented as a part of the at least one gate device 102a-102d. In the example of FIGS. 1A-1C, the control unit 104 is an external entity. The external entity herein means an entity that locates separate from the at least one gate device 102a-102d. The implementation of the control unit 104 may be done as a stand-alone control entity or as a distributed control environment between a plurality of stand-alone control entities, such as a plurality of servers providing distributed control resource. In other words, if the access control system 100 comprises more than one gate device 102a-102d, the control unit 104 may be implemented as one centralized stand-alone control entity configured to control all gate devices 102a-102d as illustrated in the examples of FIGS. 1A-1C or the control unit 104 may be implemented as a distributed control environment comprising a plurality of stand-alone control

5

entities, each configured to control one or more gate devices **102a-102d**. The control unit **104** may be communicatively coupled to the at least one gate device **102a-102d** and any other entities of the access control system **100**. The communication between the control unit **104** and the other entities of the access control system **100** may be based on one or more known communication technologies, either wired or wireless.

The invention is described next by using one gate device **102a** as illustrated in an example of FIGS. 2A and 2B. However, the invention is not limited to that and when the access control system **100** according to the invention comprises more than one gate device **102a-102d**, each gate device **102a-102d** operates as will be described by referring to the one gate device **102a**. For sake of clarity the control unit **104** is not shown in FIGS. 2A and 2B.

The control unit **104** may be configured by default to control the gate device **102a** to operate in the first operation mode, in which the gate device **102a** is arranged to allow access for the users **106** via the gate device **102a** only from one access direction, e.g. a first access direction **202** of the gate device **102a**. The allowing the access via the gate device **102a** may comprise also indicating the access via the gate device **102a** by means of the visual indication **108**, **108a**, **108b** as will be described. FIG. 2A illustrates schematically an example of the operation of the gate device **102a** in the first operation mode. The control unit **104** may detect that the traffic flow of the users **106** meets a predefined capacity limit based on traffic flow data of the users **106**. The capacity limit of the gate device **102a** may be defined by a predefined number of users **106** accessing via the gate device **102a** per a predefined time period, e.g. 30 users per one minute. In response to detecting that the traffic flow of the users **106** meets the predefined capacity limit, the control unit **104** is configured control the gate device **102a** to change to operate in the second operation mode, in which the gate device **102a** is arranged to allow access for the users via the gate device **102a** from both access directions, e.g. from the first access direction **202** and from a second access direction **204** of the gate device **102a**. The capability of operating in the first operation mode and in the second operation mode improves an operation capacity of the at least one gate device **102a-102d** of the access control system **100**. Alternatively or in addition, the capability of operating in the first operation mode and in the second operation mode improves the versatility of the at least one gate device **102a-102d** of the access control system **100**. In the second operation mode the gate device **102a** may be arranged to allow simultaneous and/or non-simultaneous access of the users via the gate device **102a** from both access directions, e.g. from the first access direction **202** and from the second access direction **204**. In other words, the users **106** are allowed to access via the gate device **102a** simultaneously from both access directions **202**, **204** or one user **106** at a time, e.g. first the user **106a** from the first access direction **202** and then the user **106b** from the second access direction **204** or vice versa. FIG. 2B illustrates schematically an example of the operation of the gate device **102a** in the second operation mode.

According to an example embodiment of the invention, when the gate device **102a** is operating in the second mode, the control unit **104** may further be configured to detect that the traffic flow of the users **106** is less than the predefined capacity limit based the traffic flow data. In response to detecting that the traffic flow of the users **106** is less than the predefined capacity limit, the control unit **104** may be

6

configured to control the gate device **102a** to change back to operate in the first operation mode.

According to an example embodiment of the invention, the access control system **100** may further comprise at least one imaging device **206** configured to obtain image data from both access directions, e.g. from the first access direction **202** and from the second access direction **204**. The at least one imaging device **206** may comprise one or more optical imaging devices, e.g. cameras. The at least one imaging device **206** may enable detection and/or identification of a user **106** at a distance away from the gate device **102a**. The distance may be e.g. between 0 to 10 meters from the gate device **102a** and preferably between 1 to 2 meters, 1 to 3 meters or 1 to 5 meters. The at least one imaging device **206** may be arranged to the gate device **102a** or in a vicinity of, i.e. close to, the gate device **102a**, e.g. to a wall, a ceiling and/or to a separate support device. In the example of FIGS. 2A and 2B the at least one imaging device **206** is arranged to a wall close to the gate device **102a** so that the at least one imaging device **206** is capable to obtain the image data from both access directions **202**, **204**.

The control unit **104** may be configured to receive the obtained image data from the first access direction **202** in the first operation mode of the gate device **102a** or from the first access direction **202** and the second access direction **204** in the second operation mode of the gate device **102a**. Furthermore, the control unit **104** may be configured to identify authorized users approaching the gate device **102a** based on the received image data and prestored data associated with the authorized users by using facial recognition-based identification. In other words, the received image data may be compared to the prestored data in order to find a possible match from the prestored data. The prestored data may be stored in a memory or a database connected to the control unit **104**, or in a cloud service accessible via a network connection. For example, if the access control system **100** is arranged in a building, image data relating to people regularly residing, e.g. working or living, in the building may have been prestored in the memory or database. The control unit **104** may be configured to control the gate device **102a** to allow access only for the authorized users via the gate device **102** in the first operating mode of the gate device **102a** and/or in the second operating mode of the gate device **102a**.

The gate device **102a** may comprise a barrier device, such as door panel(s), turnstile, boom or any other barrier device, for preventing unauthorized users without an access via the gate device **102a**. Alternatively or in addition, the gate device **102a** may comprise one or more alarm devices, e.g. visual and/or audio alarm devices, for generating an alarm, if an unauthorized user without an access via the gate device **102** attempts to pass through the gate device **102a**. The gate device **102a** may be by default at a state allowing an unrestricted access via the gate device **102a**. In other words, the gate device **102a** is maintained at the state allowing the unrestricted access via the gate device **102a**. If the received image data relates to an unauthorized user without an access via the gate device **102a**, the control unit **104** may be configured to control the barrier device to prevent, i.e. restrict, the access via the gate device **102a**, e.g. by closing the barrier device of the gate device **102a**. Alternatively or in addition, the control unit **104** may be configured to control the one or more alarm devices to generate an alarm to indicate an access of an unauthorized-user, if the received image data relates to an unauthorized user without an access via the gate device **102a**. Alternatively, the gate device **102a** may be by default at a state preventing, i.e. restricting, the

access of the users via the gate device **102a**, i.e. the gate device **102** may be in a closed state. If the received image data relates to an authorized user with an access via the gate device **102a** the control unit **104** may be configured to control the barrier device to provide the access via the gate device **102a**, e.g. by opening the barrier device of the gate device **102**.

The access control system **100** may comprise one or more sensor devices, e.g. people counter devices, for counting the number of users **106** accessing via the at least one gate device **102a-102d** in order to provide the traffic flow data. The one or more sensor devices may be based e.g. on infrared imaging, thermal imaging, video imaging, and/or wireless local area networks, e.g. Wi-Fi. The video imaging-based user counting may use algorithms, e.g. machine learning algorithms, to count the number or users **106** accessing via the at least one gate device **102a-102d** from image data obtained with a video imaging device, e.g. camera. According to an example embodiment of the invention, the at least one imaging device **206** may be used in the video imaging-based user counting. The control unit **104** may obtain the traffic flow data from the one or more sensor devices constantly, i.e. in real time, in order to obtain real-time traffic flow data. Alternatively or in addition, the traffic flow data may be based on statistics of traffic flow data gathered during a predefined term by the one or more sensor devices. Alternatively or in addition, the traffic flow data may be based on estimated traffic flow, e.g. based on time of day. For example, it may be estimated that during typical rush hour times, e.g. in the morning and/or in the afternoon, the traffic flow may be high and thus meet the predefined capacity limit of the gate device **102a**.

As discussed above, the access control system **100** may comprise at least one indication device **208a, 208b** arranged to provide a visual indication **108** indicating the access direction, e.g. the first access direction **202** and/or the second access direction **204**, depending on the operation mode of the gate device **102a**. The at least one indication device **208a, 208b** may comprise one or more displays on which the visual indication **108** may be displayed and/or one or more projectors configured to project the visual indication **108** on a surface in a vicinity of, i.e. close to, the gate device **102a**. In the example of FIG. 2A, wherein the gate device **102a** is operating in the first operating mode of the gate device **102a**, the at least one indication device **208a, 208b** comprises a first projector **208a** arranged to the gate device **102a** and configured to project the visual indication **108a** indicating the access direction, i.e. the first access direction **202**, on a surface in front of the gate device **102a** on the access direction side of the gate device **102a**, i.e. on the first access direction **202** side of the gate device **102a**. In the example of FIG. 2A the at least one indication device **208a, 208b** comprises a second projector **208b** arranged to the gate device **102a** and configured to project an optional further visual indication **108b** indicating the prohibited access direction, i.e. the second access direction **204**, on a surface in front of the gate device **102a** on the prohibited access direction side of the gate device **102a**, i.e. on the second access direction **204** side of the gate device **102a**.

In the second operation mode of the gate device **102a** the at least one indication device **208a, 208b** may be arranged to provide visual indication **108a, 108b** indicating two lanes, wherein one of the two lanes indicates the first access direction **202** and the other of the two lanes indicates the second access direction **204**. This is illustrated e.g. in the example of FIG. 2B. In the example of FIG. 2B, wherein the gate device **102a** is operating in the second operating mode

of the gate device **102a**, the at least one indication device **208a, 208b** comprises the first projector **208a** and the second projector **208b**. The first projector **208a** is arranged to the gate device **102a** and configured to project the visual indication **108a** indicating the first access direction **202** on a surface in front of the gate device **102a** on the first access direction **202** side of the gate device **102a**. The second projector **208b** is arranged to the gate device **102a** and configured to project the visual indication **108b** indicating the second access direction **204** on a surface in front of the gate device **102a** on the second access direction **204** of the gate device **102a**.

Next an example of the method according to the invention is described by referring to FIG. 3. FIG. 3 schematically illustrates the invention as a flow chart.

At a step **302**, the control unit **104** controls the gate device **102a** to operate by default in a first operation mode, in which the gate device **102a** is arranged to allow access for the users **106** via the gate device **102** only from one direction, e.g. a first access direction **202**. In other words, in the first operation mode the gate device **102a** allows one-way access via the gate device **102a**. The allowing the access via the gate device **102a** may comprise also indicating the access via the gate device **102a** by means of the visual indication **108, 108a, 108b** as will be described.

At a step **304**, the control unit **104** detects that a traffic flow of the users **106** meets a pre-defined capacity limit based on traffic flow data of the users **106**. The capacity limit of the gate device **102a** may be defined by a predefined number of users **106** accessing via the gate device **102a** per a predefined time period, e.g. **30** users per one minute.

At a step **306**, in response to detecting that the traffic flow of the users **106** meets the predefined capacity limit, the control unit **104** controls the gate device **102a** to change to operate in a second operation mode, in which the gate device **102a** is arranged to allow access for the users **106** via the gate device **102a** from the both access directions, e.g. from the first access direction **202** and from the second access direction **204**. In the second operation mode the gate device **102a** may be arranged to allow simultaneous and/or non-simultaneous access of the users **106** via the gate device **102a** from both access directions, e.g. from the first access direction **202** and from the second access direction **204**. In other words, the users **106** are allowed to access via the gate device **102a** simultaneously from both access directions **202, 204** or one user **106** at a time, e.g. first the user **106a** from the first access direction **202** and then the user **106b** from the second access direction **204** or vice versa.

According to an example embodiment of the invention, when the gate device **102a** is operating in the second mode, at a step **306** the control unit **104** may further detect that the traffic flow of the users **106** is less than the predefined capacity limit based the traffic flow data. In response to detecting that the traffic flow of the users **106a** is less than the predefined capacity limit, the control unit **104** may control the gate device **102a** to change back to operate in the first operation mode as in the step **302**.

According to an example embodiment of the invention, the access control system **100** may further comprise at least one imaging device **206** for obtaining image data from both access directions, e.g. from the first access direction **202** and from the second access direction **204**. The at least one imaging device **206** may enable detection and/or identification of a user **106** at a distance away from the gate device **102a**. The method may comprise receiving, by the control unit **104**, the image data from the first access direction **202** in the first operation mode of the gate device **102a** or from

the first access direction **202** and from the second access direction **204** in the second operation mode of the gate device **102a**. The method may further comprise identifying, by the control unit **104**, authorized users approaching the gate device **102a** based on the received image data and prestored data associated with the authorized users by using facial recognition-based identification. In other words, the received image data may be compared to the prestored data in order to find a possible match from the prestored data. The prestored data may be stored in a memory or a database connected to the control unit **104**, or in a cloud service accessible via a network connection. For example, if the access control system **100** is arranged in a building, image data relating to people regularly residing, e.g. working or living, in the building may have been prestored in the memory or database. The method may further comprise controlling, by the control unit **104**, the gate device **102a** to allow the access only for the authorized users via the gate device **102a** in the first operating mode of the gate device **102a** and/or in the second operating mode of the gate device **102a**.

According to an example embodiment of the invention, the method comprises provide the traffic flow data by counting the number of users **106** accessing via the at least one gate device **102a-102d** with one or more sensor devices, e.g. people counter devices. The one or more sensor devices may be based e.g. on infrared imaging, thermal imaging, video imaging, and/or wireless local area networks, e.g. Wi-Fi. The video imaging-based user counting may use algorithms, e.g. machine learning algorithms, to count the number or users **106** accessing via the at least one gate device **102a-102d** from image data obtained with a video imaging device, e.g. camera. According to an example of the invention, the at least one imaging device **206** may be used in the video imaging-based user counting. The control unit **104** may obtain the traffic flow data from the one or more sensor devices constantly, i.e. in real time, in order to obtain real-time traffic flow data. Alternatively or in addition, the traffic flow data may be based on statistics of traffic flow data gathered during a predefined term by the one or more sensor devices. Alternatively or in addition, the traffic flow data may be based on estimated traffic flow, e.g. based on time of day. For example, it may be estimated that during typical rush hour times, e.g. in the morning and/or in the afternoon, the traffic flow may be high and thus meet the predefined capacity limit of the gate device **102a**.

According to an example embodiment of the invention, the method may further comprise providing, by at least one indication device **208a, 208b**, a visual indication **108** indicating the access direction, e.g. the first access direction **202** and/or the second access direction **204**, depending on the operation mode of the gate device **102a**. The visual indication **108, 108a, 108b** may be displayed on one or more displays and/or projected on a surface in a vicinity of, i.e. close to, the gate device **102a**.

According to an example embodiment of the invention, the method may comprise providing, by at least one indication device **208a, 208b**, in the second operation mode a visual indication **108a, 108b** indicating two lanes, wherein one of the two lanes indicates the first access direction **202** and the other of the two lanes indicates the second access direction **204**.

FIG. 4 schematically illustrates an example of components of the control unit **104** according to the invention. The control unit **104** may comprise a processing unit **410** comprising one or more processors, a memory unit **420** comprising one or more memories, a communication unit **430**

comprising one or more communication devices, and possibly a user interface (UI) unit **440**. The memory unit **420** may store portions of computer program code **425** and any other data, and the processing unit **410** may cause the control unit **104** to operate as described by executing at least some portions of the computer program code **425** stored in the memory unit **420**. The communication unit **430** may be based on at least one known communication technologies, either wired or wireless, in order to exchange pieces of information as described earlier. The communication unit **430** provides an interface for communication with any external unit, such as the at least one gate device **102a-102d**, the at least one imaging device **206**, the at least one indication device **208a, 208b**, database and/or any external entities or systems. The communication unit **430** may comprise one or more communication devices, e.g. radio transceiver, antenna, etc. The user interface **440** may comprise I/O devices, such as buttons, key-board, touch screen, microphone, loudspeaker, display and so on, for receiving input and outputting information. The computer program **425** may be stored in a non-statutory tangible computer readable medium, e.g. an USB stick or a CD-ROM disc.

The verb "meet" in context of a predefined capacity limit is used in this patent application to mean that a predefined condition is fulfilled. For example, the predefined condition may be that the predefined capacity limit is reached and/or exceeded.

The specific examples provided in the description given above should not be construed as limiting the applicability and/or the interpretation of the appended claims. Lists and groups of examples provided in the description given above are not exhaustive unless otherwise explicitly stated.

The invention claimed is:

1. An access control system comprising:
at least one gate device, and
a control unit configured to:

control the at least one gate device to operate in a first operation mode, in which the gate device is arranged to allow access for the users via the gate device only from a first access direction,
detect that a traffic flow of the users meets a predefined capacity limit,

control at least one gate device to change to operate in a second operation mode, in which the gate device is arranged to allow access for the users via the gate device from the first access direction and from the second access direction based on the detected traffic flow of the users meeting the predefined capacity limit,

control the gate device, in the second mode, to close and act as a barrier restricting access in the first access direction and the second access direction in response to an unauthorized user being detected as approaching the gate device, and

control the gate device, in the second mode, to remain open to allow access in both the first direction and the second direction simultaneously in response to no unauthorized users being detected as approaching the gate device.

2. The access control system according to claim 1, further comprising at least one imaging device configured to obtain image data from the first access direction and from the second access direction, wherein the control unit is configured to:

receive the obtained image data from the first access direction in the first operation mode of the gate device

11

or from the first access direction and the second access direction in the second operation mode of the gate device,

identify authorized users approaching the at least one gate device based on the received image data and prestored data associated with the authorized users by using facial recognition-based identification, and

control the at least one gate device to allow the access only for the authorized users via the at least one gate device.

3. The access control system according to claim 1, wherein the control unit is further configured to:

detect that the traffic flow of the users is less than the predefined capacity limit, and

control the at least one gate device to change back to operate in the first operation mode.

4. The access control system according claim 1, wherein in the second operation mode the gate device is arranged to allow simultaneous access of the users via the gate device from the first access direction and from the second access direction.

5. The access control system according to claim 1, wherein the capacity limit of the at least one gate device is defined by a predefined number of users accessing via the at least one gate device per a predefined time period.

6. The access control system according claim 1 further comprising at least one indication device arranged to provide a visual indication indicating the first access direction and/or the second access direction depending on the operation mode of the gate device.

7. The access control system according to claim 6, wherein in the second operation mode the at least one indication device is arranged to provide visual indication indicating two lanes, wherein one of the two lanes indicates the first access direction and the other of the two lanes indicates the second access direction.

8. The access control system according to claim 6, wherein the at least one indication device comprises one or more displays on which the visual indication is displayed and/or one or more projectors configured to project the visual indication on a surface in a vicinity of the at least one gate device.

9. A method for controlling operation of an access control system comprising at least one gate device and a control unit, the method comprises:

controlling, by the control unit, the at least one gate device to operate in a first operation mode, in which the gate device is arranged to allow access for the users via the gate device only from a first access direction,

detecting, by the control unit, that a traffic flow of the users meets a predefined capacity limit,

controlling, by the control unit, at least one gate device to change to operate in a second operation mode, in which the gate device is arranged to allow access for the users via the gate device from the first access direction and from a second access direction based on the detected traffic flow of the users meeting the predefined capacity limit,

controlling, by the control unit, the gate device, in the second mode, to close and act as a barrier restricting access in the first access direction and the second access direction in response to an unauthorized user being detected as approaching the gate device, and

controlling, by the control unit, the gate device, in the second mode, to remain open to allow access in both the first direction and the second direction simultane-

12

ously in response to no unauthorized users being detected as approaching the gate device.

10. The method according to claim 9, wherein the system further comprises at least one imaging device configured to obtain image data from the first access direction and from the second access direction, wherein method further comprising:

receiving, by the control unit, the image data from the first access direction in the first operation mode of the gate device or from the first access direction and the second access direction in the second operation mode of the gate device,

identifying, by the control unit, authorized users approaching the at least one gate device based on the received image data and prestored data associated with the authorized users by using facial recognition-based identification, and

controlling, by the control unit, the at least one gate device to allow the access only for the authorized users via the at least one gate device.

11. The method according to claim 9, wherein the method further comprises:

detecting, by the control unit, that the traffic flow of the users is less than the predefined capacity limit, and

controlling, by the control unit, the at least one gate device to change back to operate in the first operation mode.

12. The method according to claim 9, wherein in the second operation mode the gate device is arranged to allow simultaneous access of the users via the gate device from the first access direction and from the second access direction.

13. The method according to claim 9, wherein the capacity limit of the at least one gate device is defined by a predefined number of users accessing via the at least one gate device per a predefined time period.

14. The method according to claim 9, wherein the method further comprises providing a visual indication indicating the first access direction and/or the second access direction depending on the operation mode of the gate device.

15. The method according to claim 14, wherein the method further comprises providing in the second operation mode a visual indication indicating two lanes, wherein one of the two lanes indicates the first access direction and the other of the two lanes indicates the second access direction.

16. The method according to claim 14, wherein the visual indication is displayed on one or more displays and/or projected on a surface in a vicinity of the at least one gate device.

17. The access control system according to claim 2, wherein the control unit is further configured to:

detect that the traffic flow of the users is less than the predefined capacity limit, and

control the at least one gate device to change back to operate in the first operation mode.

18. The access control system according claim 2, wherein in the second operation mode the gate device is arranged to allow simultaneous access of the users via the gate device from the first access direction and from the second access direction.

19. The access control system according claim 3, wherein in the second operation mode the gate device is arranged to allow simultaneous access of the users via the gate device from the first access direction and from the second access direction.

20. The access control system according to claim 2, wherein the capacity limit of the at least one gate device is

13

defined by a predefined number of users accessing via the at
least one gate device per a predefined time period.

* * * * *

14